1. Towers of Hanoi

Constraint: Moves allowed only $A \leftrightarrow B$ and $B \leftrightarrow C$ (not $A \leftrightarrow C$).

$$\begin{aligned} M(n) &= \text{moves for } A \to C \text{ (non-adjacent)} \\ M'(n) &= \text{moves for } A \to B \text{ (adjacent)} \end{aligned}$$

Algorithm for $A \to C$:

1. Move $(n-1)$ disks: $A \to C$ $[M(n-1)]$
2. Move disk n: $A \to B$ [1]
3. Move $(n-1)$ disks: $C \to A$ $[M(n-1)]$
4. Move disk n: $B \to C$ [1]
5. Move $(n-1)$ disks: $A \to C$ $[M(n-1)]$

$$\begin{aligned} M(n) &= 3M(n-1) + 2, \quad M(1) = 2 \\ M(n) &= A \cdot 3^n - 1 \\ M(1) = 2 &\implies 3A - 1 = 2 \implies A = 1 \end{aligned}$$

$$\boxed{M(n) = 3^n - 1}$$

2. Partition

(a)

```
begin
    PARTITION(A[1..n], p)
    k ⟵ 0
    for i ⟵ 1 to n do
        if A[i] ≤ p then then
            k ⟵ k + 1
            SWAP(A[k], A[i])
    return k
```

Start of each loop iteration:

$A[1..k] \le p$

$A[k+1..i-1] > p$

When the algorithm ends:

$A[1..k] \le p$

$A[k+1..n] > p$

(b)

Comparisons: n

Swaps: $\le n$

$$\boxed{T(n) = O(n)}$$

3. Selection Recurrence

(a) Groups of size 3:

$$\begin{aligned}
\text{Groups:}\quad & n/3 \\
\text{Elements} \leq \text{group median:}\quad & 2 \\
\text{Medians} \leq \text{pivot:}\quad & \geq \frac{1}{2} \cdot \frac{n}{3} = \frac{n}{6} \\
\text{Guaranteed} \leq \text{pivot:}\quad & 2 \cdot \frac{n}{6} = \frac{n}{3} \\
\text{Worst-case recurse:}\quad & n - \frac{n}{3} = \frac{2n}{3}
\end{aligned}$$

$$\boxed{T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + O(n)}$$

Note: $\frac{1}{3} + \frac{2}{3} = 1 \implies T(n) = \Theta(n \log n)$

(b) Groups of size $2k+1$:

$$\begin{aligned}
\text{Groups:}\quad & \frac{n}{2k+1} \\
\text{Elements} \leq \text{group median:}\quad & k+1 \\
\text{Medians} \leq \text{pivot:}\quad & \geq \frac{n}{2(2k+1)} \\
\text{Guaranteed} \leq \text{pivot:}\quad & (k+1) \cdot \frac{n}{2(2k+1)} = \frac{(k+1)n}{2(2k+1)}
\end{aligned}$$

$$\begin{aligned}
\text{Worst-case:}\quad n - \frac{(k+1)n}{2(2k+1)} &= n \cdot \frac{2(2k+1) - (k+1)}{2(2k+1)} \\
&= n \cdot \frac{3k+1}{2(2k+1)}
\end{aligned}$$

$$\boxed{T(n) = T\left(\frac{n}{2k+1}\right) + T\left(\frac{(3k+1)n}{2(2k+1)}\right) + O(n)}$$

Linear time condition:

$$\frac{1}{2k+1} + \frac{3k+1}{2(2k+1)} = \frac{3(k+1)}{2(2k+1)} < 1 \iff k > 1$$

4. Intersection Detection

begin

$\quad$ IntersectionDetect($I[1..n]$)

$\quad$ **Sort** I by $A[i]$

$\quad$ $m \longleftarrow B[1]$

$\quad$ **for** $i \longleftarrow 2$ ***to*** n **do**

$\quad\quad$ **if** $A[i] < m$ ***then*** **then**

$\quad\quad\quad$ **return** ***true***

$\quad\quad$ $m \longleftarrow \max(m, B[i])$

$\quad$ **return** ***false***

$$\forall j < i : A[j] \leq A[i]$$

$$\text{overlap} \iff A[i] < B[j]$$

$T(n) = 2T(n/2) + O(n)$

Sorting: $\quad O(n \log n)$

Scan: $\quad O(n)$

$$\boxed{T(n) = O(n \log n)}$$